## Assignment 1: (Table Creation and Single Table Retrieval)

### A. Table creation without constraints:

**a.** Customer Master table : **cust**

| Column Name | Data Type |
|---|---|
| cust_id | varchar2(5) |
| lname | varchar2(15) |
| fname | varchar2(15) |
| Area | varchar2(10) |
| phone_no | number(10) |

**Records:**

| cust_id | Lname | fname | Area | phone_no |
|---|---|---|---|---|
| a01 | Patel | Ajay | Anand | 381334 |
| a02 | Shah | Anjali | Surat | 556037 |
| a03 | Thakkar | Vinay | Baroda | 372631 |
| a04 | Joshi | Kamal | Surat | 666612 |
| a05 | Pandya | Rohan | Anand | |
| a06 | Patel | Kajal | Baroda | 512527 |

**b.** Movies Master table: **movie**

| Column Name | Data Type |
|---|---|
| mv_no | number(5) |
| Title | varchar2(25) |
| Type | varchar2(10) |
| Star | varchar2(25) |
| Price | number(8,2) |

**Records:**

| mv_no | Title | type | star | price |
|---|---|---|---|---|
| 1 | Hera Pheri | Comedy | Paresh Raval | 105 |
| 2 | Kuch Kuch Hota Hein | Romantic | Kajol | 225.00 |
| 3 | Drashyam | Suspense | Ajay Devagan | 250 |
| 4 | Welcome | Comedy | Anil Kapoor | 200.00 |
| 5 | Gupt | Suspense | Bobby Deol | 195 |
| 6 | Tere Naam | Romantic | Salman Khan | 190 |
| 7 | Hungama | Comedy | Akshay Khanna | 135 |

| | | | | |
|---|---|---|---|---|
| 8 | Ajanabee | Suspense | Akshay Kumar | 165 |
| 9 | No Entry | Comedy | Anil Kapoor | 220 |
| 10 | Singham | Action | Ajay Devagan | 195 |

| B. | Single Table Retrieval |
|---|---|
| 1 | Display entire content of cust table. |
| 2 | Display customer first name with their area. |
| 3 | Display customer with their fname and last name whose cust_id is a01 or a02. |
| 4 | Display all the records from cust table whose cust_id is a02 and area is Surat. |
| 5 | Display customer details who does not have phone number. |
| 6 | Display entire content of movie table. |
| 7 | Display all details of movie whose price is greater than 200. |
| 8 | Display title and type of movie whose price is greater than 150 and less than 200. |
| 9 | Display movie details whose movie type is Comedy or Action.(Using IN and OR operator) |
| 10 | Display movie details whose movie type is Suspense and price is greater than 180. |
| 11 | Display title, star and price of comedy movie. |
| 12 | Display distinct movies types. |
| 13 | Display movies details in ascending order of movie title. |
| 14 | Display movies details in descending order of movie price. |
| 15 | Display movie details whose type is Romantic or Action and price less than 200. |

## Assignment 2: (Table Creation with Constraints)

**TABLE NAME: dept**

| Column Name | Data Type | Constraints |
|---|---|---|
| Deptno | number(2) | Primary Key |
| Dname | varchar2(15) | |
| Loc | varchar2(15) | |

**Records :**

| deptno | dname | loc |
|---|---|---|
| 10 | Accounting | New York |
| 20 | Research | Dallas |
| 30 | Sales | Chicago |
| 40 | Operations | Boston |

**TABLE NAME: emp**

| Column Name | Data Type | Constraints |
|---|---|---|
| Empno | number(4) | Primary Key |
| Ename | varchar2(10) | |

| | | |
|---|---|---|
| Job | varchar2(10) | |
| Mgr | number(4) | |
| Hiredate | date | |
| Sal | number(7,2) | |
| Comm. | number(7,2) | |
| Deptno | number(2) | Reference to dept table |

**Records:**

| empno | ename | Job | mgr | hiredate | sal | comm | deptno |
|---|---|---|---|---|---|---|---|
| 7369 | Smith | Clerk | 7902 | 17-DEC-80 | 800 | | 20 |
| 7499 | Allen | Salesman | 7698 | 20-FEB-81 | 1600 | 300 | 30 |
| 7521 | Ward | Salesman | 7698 | 22-FEB-81 | 1250 | 500 | 30 |
| 7566 | Jones | Manager | 7839 | 02-APR-81 | 2975 | | 20 |
| 7654 | Martin | Salesman | 7698 | 28-SEP-81 | 1250 | 1400 | 30 |
| 7698 | Blake | Manager | 7839 | 01-MAY-81 | 2850 | | 30 |
| 7782 | Clark | Manager | 7839 | 09-JUN-81 | 2450 | | 10 |
| 7788 | Scott | Analyst | 7566 | 09-DEC-82 | 3000 | | 20 |
| 7839 | King | President | | 17-NOV-81 | 5000 | | 10 |
| 7844 | Turner | Salesman | 7698 | 08-SEP-81 | 1500 | 0 | 30 |
| 7876 | Adams | Clerk | 7788 | 12-JAN-83 | 1100 | | 20 |
| 7900 | James | Clerk | 7698 | 03-DEC-81 | 950 | | 30 |
| 7902 | Ford | Analyst | 7566 | 03-DEC-81 | 3000 | | 20 |
| 7934 | Miller | Clerk | 7782 | 23-JAN-82 | 1300 | | 10 |

**TABLE NAME: doctor**

| Column Name | Data Type | Constraints |
|---|---|---|
| d_id | varchar2(3) | Primary Key, Must start with D |
| dname | varchar2(25) | |
| DCity | varchar2(20) | Must be ‘Anand’,’Baroda’ or ‘Nadiad’ |
| spec | varchar2(20) | Must be in ‘ENT’, ‘Dental’,’Ortho’ and ‘Eye’ |

**Records :**

| D_id | Dname | DCity | Spec |
|---|---|---|---|
| D01 | Dharmendra Patel | Anand | ENT |
| D02 | Jinay Shah | Nadiad | Eye |
| D03 | Sonal Patel | Anand | Dental |
| D04 | Mehul Patel | Anand | ENT |
| D05 | Mahesh Patel | Baroda | Ortho |
| D06 | Neerav Parmar | Nadiad | Dental |
| D07 | Niraj Shah | Anand | Ortho |
| D08 | Mukesh Kotiya | Baroda | Eye |

**TABLE NAME: patient**

| Column Name | Data Type | Constraints |
|---|---|---|
| P_id | varchar2(3) | Primary Key, Must start with P |
| Pname | varchar2(25) | |
| Bdate | date | Less than system date |
| Paddr | Varchar2(35) | |
| Gender | Char(1) | Check either ‘F’ or ‘M’, default ‘M’; |

**Records :**

| p_id | Pname | Bdate | paddr | Gender |
|---|---|---|---|---|
| P01 | Rupal Patel | 12-SEP-98 | Anand | F |
| P02 | Kalpesh Shah | 04-SEP-99 | Baroda | M |
| P03 | Kavita Joshi | 23-OCT-87 | Surat | F |
| P04 | Sachin Pandya | 04-FEB-89 | Anand | M |
| P05 | Brijisha Patel | 15-MAR-95 | Nadiad | F |
| P06 | Mayank Shah | 16-MAY-65 | Baroda | M |
| P07 | Mitul Soni | 27-AUG-96 | Anand | M |
| P08 | Krunal Pandya | 20-JAN-70 | Rajkot | M |
| P09 | Rinkal Bhatt | 14-APR-95 | Navsari | M |
| P10 | Pratik Solanki | 22-JUN-87 | Surat | M |

**TABLE NAME: treatment**

| Column Name | Data Type | Constraints |
|---|---|---|
| T_id | varchar2(3) | Primary Key, Must start with T |
| Tdetails | varchar2(50) | |
| Charges | Number(5) | Must be greater than 0 |

**Records :**

| T_id | Tdetails | Charges |
|---|---|---|
| T01 | Retina | 1200 |
| T02 | Root Canal | 400 |
| T03 | Sleep Apnea | 600 |
| T04 | Glaucoma | 1500 |
| T05 | Tonsil | 200 |
| T06 | Knee Replacement | 500 |
| T07 | Ear Infection | 400 |
| T08 | Spine | 600 |
| T09 | Tooth Extraction | 1000 |

**TABLE NAME: appointment**

| Column Name | Data Type | Constraints |
|---|---|---|
| A_id | Number(6) | Pimary Key |
| d_id | varchar2(3) | References to doctor table |
| P_id | varchar2(3) | References to patient table |
| T_id | varchar2(3) | References to treatment table |
| App_date | Date | |
| Duration | Number(4) | |
| Status | Char(1) | Either 'C' for Confirmed or 'W' for Waiting |

**Records:**

| A_id | d_id | P_id | T_id | App_date | Duration | Status |
|---|---|---|---|---|---|---|
| 201801 | D02 | P01 | T01 | 01-DEC-18 | 20 | C |
| 201802 | D03 | P02 | T02 | 12-OCT-18 | 35 | C |
| 201803 | D04 | P03 | T03 | 26-DEC-18 | 30 | W |
| 201804 | D08 | P04 | T04 | 01-JAN-18 | 25 | C |
| 201805 | D01 | P05 | T05 | 11-SEP-18 | 40 | C |
| 201806 | D05 | P06 | T06 | 23-SEP-18 | 45 | C |
| 201807 | D04 | P07 | T07 | 16-JAN-18 | 55 | W |
| 201808 | D07 | P08 | T08 | 12-NOV-18 | 40 | W |
| 201809 | D06 | P09 | T09 | 16-NOV-18 | 35 | C |
| 201810 | D04 | P10 | T03 | 03-JUL-18 | 30 | W |
| 201811 | D01 | P04 | T05 | 01-DEC-18 | 50 | C |
| 201812 | D07 | P07 | T08 | 30-OCT-18 | 25 | W |

**Assignment-3**

| | **Single Table Retrieval for emp table** |
|---|---|
| 1 | Display employee number and name from emp table. |
| 2 | Display job from emp table, removing duplicate values. |
| 3 | Display employees of which are not in department 20. |
| 4 | Display employees, whose name begins with 'J'. |
| 5 | Display all employees who receive commission. |
| 6 | Display employees of department 10 and who hired after 1st Sep '81. |
| 7 | Display employees who are not in department 20.<br>(Use all the alternatives) |
| 8 | Display employee details like name, job and salary with user defined heading for all. |
| 9 | List employee name, salary, PF, HRA, DA & GROSS order the result in ascending order of salary.(PF = 10% salary, HRA=50%, DA=30%,GROSS=SAL+HRA+DA-PF) |
| 10 | Display names of all the employees along with their annual salary. |
| 11 | Display employees, whose name begins with 'J'. |
| 12 | Display all employees whose name begins with 'J' and has 'N' as the 3rd character. |
| 13 | Display emp details whose job is Clerk, Salesman or Analyst. |

| | |
|---|---|
| 14 | Display emp details whose name begins with ‘J’ and job is ‘Clerk’. |
| 15 | Display emp details in descending order of their name. |

## Assignment-4

| | **Queries using Functions** |
|---|---|
| | |
| 1. | Display empno, ename in uppercase and salary form emp table. |
| 2 | Display empno, ename in lower case and job form emp table whose sal>300 |
| 3 | Display maximum salary and minimum salary from emp table. |
| 4 | Display total salary and average salary from emp table whose deptno is 10. |
| 5 | Display total no of employee working in deptno 20 and 30. |

## Assignment-5

| | **Queries using Date.** |
|---|---|
| 1 | Display empno,enameand salary with hiredate. The format of hiredate is 12th Jan,1981. |
| 2 | Display empno,enameand job and salary with hiredate. The format of hiredate is twelveth January,81. |
| 3 | Display empno,ename and job and salary who hired in the month of September. |
| 4 | Display empno,ename and commision who hired on Monday |
| 5 | Find the date,15 days after today’s date. |
| 6 | Display all records from emp table whose working experience is more than 20 years. |

## Assignment-6

| | **Group by…having** |
|---|---|
| 1 | Display total no. of employee working is each department. |
| 2 | Find total no. of salesman and clerk in each department. |
| 3 | Find maximum salary of emp in each department. |
| 4 | Display dname and average salary of each department. |
| 5 | Display jobwise commission. |
| 6 | Display total salary of clerk and analyst of each department. |
| 7 | Display ename , dnmae and deptno of employee whose maximum salary greater than 5000. |
| 8 | Display deptno and salary whose department have more than 2 employee |
| 9 | Display deptno and salary whose total salary of employee is lessthan 2000. |
| 10 | Display total no of patient citywise. |
| 11 | Display total no. of doctors for each specialization. |
| 12 | Display total no of male patient citywise. |
| 13 | Display total no. of doctors for eye specialization. |
| | |

## Assignment-7

| | **Joins** |
|---|---|

| | |
|---|---|
| 1 | Display ename and dname of employee who working in Accounting department. |
| 2 | Display deptno , dname and ename of employee in sales department whose sal>2000 |
| 3 | Display ename, dname of emp in research department whose enmae start with ‘A’. |
| 4 | Display ename, dname and sal of emp working as ‘clerk’ or ‘salesman’. |
| 5 | Display dname, ename,deptn of emp hired before ’01-sep-81’, |
| 6 | Display treatment details with their charges. |
| 7 | Display patients name and their appointment date. |
| 8 | Display doctor name, specialization and their appointment date. |
| 9 | Display treatment details whose charges is greater than 500. |
| 10 | Display patients name and their appointment date whose appointment date in the month of December. |
| 11 | Display doctor name , patient name and treatment details with their charges. |
| 12 | Display detailed list of appointments for dr. Dharmendra Patel. |
| 13 | Display patient name who was treted by Dr. Mehul Patel. |
| 14 | Display patient name whose treatment is not completed. |
| 15 | Display the age of all male patients who are presently undergoing treatment. |
| 16 | Display employee name with their manage name. |

**Assignment-8**

| | **Group by….having (Multiple Tables)** |
|---|---|
| 1 | Display deptno, dname and total no. of employee for each department. |
| 2 | Display dname,total salary paid to all clerk for each department. |
| 3 | Display dname, minimum salary paid to all salesman for each department whose minimum salary>400. |
| 4 | Display the total amount received by Dr. Mehul Patel. |
| 5 | Display all the doctors who are presently treating at least two patient. |
| 6 | Display total no. of patient treated by each doctor. |

**Assignment-9**

| | **Sub query** |
|---|---|
| 1 | Display empno, ename and salary whose salary is same as ‘WARD’s salary. |
| 2 | Display empno, ename and job who are working in same as ‘Scott’. |
| 3 | List the treatment details which treated in anand. |
| 4 | Display patient name with age greater than 28 years and completed their treatment. |
| 5 | Display doctor name who have treated male patients. |

**Assignment 10 (View)**

| **Table structure:** | |
|---|---|
| **Dept11 (dept_id number(5) primary key, dept_name varchar2(20) not null)** | **Emp11(emp_id number(3) primary key, emp_name varchar2(20) not null, emp_sal number(8,2), dept_id number(5) references dept)** |

| | |
|---|---|
| 1 | WAQ to create a view having name “emp_vw” which contains emp_id, emp_name and salary from emp table. |
| 2 | WAQ to add new records in “emp” table using “emp_vw” view. |
| 3 | WAQ to update salary of JOHN from 2000 to 5000. |
| 4 | WAQ to create a read only view having name “emp_ro_vw” which contains emp_id, emp_name and salary. |
| 5 | WAQ to view having name which contains emp_name, emp_sal and dept_name from dept and emp table. |

**Assignment 11 (sequence)**

| | |
|---|---|
| 1 | Create sequence **“seq1to10”** which is start with 1 and ends with 10. |
| 2 | Create sequence **“seq_even”** which is generates with even numbers between 100 and 500. |
| 3 | Create sequence **“seq_rev”** in a way that produce Z-A repetitive. |
| 4 | Create sequence **“seq_div3”**, it generates the number which is divisible by 3 and numbers are between 3 and 23. |
| 5 | Create sequence **“seq_5”** which is generates the numbers between 5 and 250, it start with 5 increment by 5, end with 250, cache value 5 and also the numbers are in order. |